

Preclass Learning: Loops and Iteration

P2 (ICA 12)

1. Functional Programming Paradigm

- **Focus:** Emphasizes writing code using expressions and declarations rather than state and mutable data. Functions are treated as first-class citizens.
- **Goal:** Cleaner code, less reliance on mutable variables, and enhanced readability.

2. The map() Family (purrr)

The map() functions apply a function to every element of an input list or vector.

Function	Output Type	Description
map()	Always returns a list .	The most generic map function.
map_dbl()	Returns an atomic double vector.	Used when the function returns a single numeric result.
map_chr()	Returns an atomic character vector.	Used when the function returns a single string result.
map_df()	Returns a data frame (or tibble).	Ideal for reading multiple files or combining results of repeated analysis.

3. Specialized Functional Tools

- **walk():** Used when the function being applied has a **side effect** (e.g., printing a message, saving a file) and you don't need the return value. It returns the input invisibly.
- **reduce():** Iteratively combines the elements of a vector or list using a two-argument function. Useful for calculating running totals or products.
- **Anonymous Functions (~):** A concise shorthand syntax in purrr for defining a simple function without naming it (e.g., `~ .x * 2`).